

PHY 130A

Lecture Notes:

The Dirac Equation

Michael Mulhearn

March 12, 2026

1 The Four Gradient and Four-Momentum Operator

In vector calculus, we define the gradient of a scalar-valued function $\phi(x, y, z)$ by:

$$\nabla\phi \equiv i\frac{\partial\phi}{\partial x} + j\frac{\partial\phi}{\partial y} + k\frac{\partial\phi}{\partial z} \equiv e_1\frac{\partial\phi}{\partial x^1} + e_2\frac{\partial\phi}{\partial x^2} + e_3\frac{\partial\phi}{\partial x^3}$$

which we have written in two notations, so that you can see the components are simply:

$$\frac{\partial\phi}{\partial x^i}$$

Similarly, for a scalar valued function $f(x)$ of a four-vector x with components x^μ we define the **four-gradient** by its components:

$$\partial_\mu f \equiv \frac{\partial f}{\partial x^\mu}$$

In the exercise, you will show that $\partial_\mu f$ transforms like the covariant components of a four vector, justifying our notation (using lowered index) and establishing the four-gradient of a scalar function as a four-vector.

Just as we write ∇ as an operator without reference to a specific function (as in $\nabla\phi$), we can likewise write:

$$\partial_\mu \equiv \frac{\partial}{\partial x^\mu}$$

Exercise 1: (*) The components x^μ of the four-vector x transform as:

$$x^{\mu'} = \Lambda^{\mu'}_\nu x^\nu$$

and

$$x^\nu = \Lambda^\nu_{\mu'} x^{\mu'}$$

For any scalar function $f(x)$, show that the components $\partial_\mu f$ of the four-gradient of f transform as:

$$\partial_\mu f = \Lambda^{\nu'}_\mu \partial_{\nu'} f$$

Hint: Use the chain rule, and note that:

$$\frac{\partial x^\mu}{\partial x^{\nu'}} = \Lambda^\mu_{\nu'}$$

□

Exercise 2: Let $\mathcal{B}$ be the standard basis for the four-vector space, so e.g.:

$$[x]_{\mathcal{B}} = \begin{pmatrix} x^0 \\ x^1 \\ x^2 \\ x^3 \end{pmatrix}$$

and $\mathcal{B}^*$ is as always the corresponding dual basis, so e.g.:

$$[x]_{\mathcal{B}^*} = \begin{pmatrix} x_0 \\ x_1 \\ x_2 \\ x_3 \end{pmatrix}$$

The covariant components ∂_μ of the four-gradient, when written as a column vector in $\mathcal{B}^*$ are

$$[\partial]_{\mathcal{B}^*} = \begin{pmatrix} \frac{\partial}{\partial t} \\ \frac{\partial}{\partial x} \\ \frac{\partial}{\partial y} \\ \frac{\partial}{\partial z} \end{pmatrix} = \begin{pmatrix} \frac{\partial}{\partial t} \\ \nabla \end{pmatrix}$$

Show that the contravariant components ∂^μ of the four-gradient, when written as a column vector in the standard basis $\mathcal{B}$ are:

$$[\partial]_{\mathcal{B}} = \begin{pmatrix} \frac{\partial}{\partial t} \\ -\nabla \end{pmatrix}$$

Hint: use the matrix representation of the metric tensor G . □

Exercise 3: In Non-Relativistic Quantum Mechanics we replace the classical quantities total energy and momentum with operators:

$$E \rightarrow \hat{E} \equiv i\hbar \frac{\partial}{\partial t}, \quad \vec{p} \rightarrow \hat{\vec{p}} \equiv -i\hbar \nabla$$

Following this same prescription, show that the four-momentum operator is:

$$\hat{p}^\mu = i\hbar \partial^\mu$$

and

$$\hat{p}_\mu = i\hbar \partial_\mu$$

2 Klein-Gordon Equation

One approach to developing a relativistic quantum mechanics is to start with the relativistic energy equation:

$$E^2 = p^2 + m^2$$

which can be written in terms of a four-vector squared as:

$$p^\mu p_\mu - m^2 = 0$$

And make the replacement:

$$p_\mu \rightarrow i\hbar\partial_\mu$$

The result is the Klein-Gordon equation:

$$-\hbar^2\partial_\mu\partial^\mu\psi - m^2\psi = 0$$

or equivalently:

$$-\frac{\partial^2\psi}{\partial t^2} + \nabla^2\psi = \left(\frac{m}{\hbar}\right)^2\psi$$

Exercise 4: Show that

$$\psi = A \sin\left(\frac{Et}{\hbar}\right)$$

is a solution to KG equation, provided $E = \pm m$. $\square$

Exercise 5: Show that $|\psi|^2$ for the previous exercise cannot possibly represent a probability density, by sketching it as function of time. $\square$

The existence of negative energy solutions (as shown in the exercises) is not a fatal problem (as we shall see). The problem is that $|\psi|^2$ integrated over all space is not conserved.

Exercise 6: Show that

$$\partial_\mu(\psi^*\partial^\mu\psi - \psi\partial^\mu\psi^*) = 0$$

$\square$

Exercise 7: When

$$\partial_\mu j^\mu = 0$$

we say that j^μ is a **conserved-current**. Show that for a conserved current j^μ then:

$$\frac{d}{dt} \int j^0 d^3x = 0$$

or equivalently:

$$Q \equiv \int j^0 d^3x$$

is a constant, which we call the conserved charge. (Hint: assume the conserved current vanishes at infinity.) $\square$

The exercises show that for:

$$\rho \equiv i(\psi^*\partial^\mu\psi - \psi\partial^\mu\psi^*)$$

we could normalize the wave function such that

$$\int \rho d^3x = 1$$

always. However, we cannot interpret ρ as a probability density because it is not positive definite, as the next exercise will show.

Exercise 8: (*) Show that:

$$\psi = A \exp(-iEt/\hbar)$$

is a solution to the KG equation if $E = \pm m$, and in this case: $\rho = 2E|\psi|^2$ which is less than zero for $E = -m$. $\square$

This renders the K-G equation unsuitable for our current purposes of finding a single particle wave function consistent with special relativity. However, in Quantum Field Theory, the notion of a single particle with positive definite probability density is abandoned, and the K-G equation emerges again as a perfectly valid equation for describing spin-0 particles.

Historically, Schrodinger first derived the K-G equation, but abandoned it when it failed to accurately predict the spectrum of the Hydrogen atom. This is because the electron in the Hydrogen atom is a spin-1/2 particle, and is not described by the KG. The (perceived) shortcomings of the K-G equation arise from the fact that it is quadratic in energy, which leads to a second order derivative in time. And this led Dirac to seek an alternative approach.

3 Dirac's Inspiration

For what is coming next let's first consider the brave mathematicians who boldly asserted that one could take the square root of negative number, by assuming the existence of the imaginary unit i with the defining property:

$$i^2 = -1$$

and carefully following that assertion through to all of its logical consequences. They discovered the complex numbers, complex algebra, and brought us an indispensable new tool. Imagine trying to do Quantum Mechanics without complex numbers! The complex numbers also gave us profound new insights. Something very similar is about to happen here!

The negative energy solutions to the Klein-Gordon equation result from the square of the energy in the relativistic energy equation:

$$E^2 = p^2 + m^2$$

Non-relativistic Quantum Mechanics has no such difficulties because the energy equation is:

$$E = \frac{p^2}{2m}$$

Dirac considered the equation:

$$E = \sqrt{p^2 + m^2}$$

and thought that we must be able to replace p with an operator and still take that square root. When he followed that through to its logical conclusions, he discovered quantum field theory. Our approach, in the next section, will be based on factoring the quadratic relativistic energy equation: So if Dirac's square root is α , we will be factoring:

$$0 = E^2 - \alpha^2 = (E - \alpha)(E + \alpha)$$

4 Dirac Equation

Following Dirac, we are attempting to factor the equation:

$$0 = p^\mu p_\mu - m^2$$

by finding appropriate γ^μ and β^μ such that:

$$p^\mu p_\mu - m^2 = (\beta^\mu p_\mu + m)(\gamma^\omega p_\omega - m)$$

Calculating:

$$(\beta^\mu p_\mu + m)(\gamma^\omega p_\omega - m) = \beta^\mu \gamma^\omega p_\mu p_\omega - m^2 + m(\gamma^\omega p_\omega - \beta^\mu p_\mu)$$

When dealing with the last term, note that we are free to adjust the dummy indices so that:

$$\gamma^\omega p_\omega - \beta^\mu p_\mu = (\gamma^\mu - \beta^\mu) p_\mu$$

which amounts to combining terms with the same indices from two different sums. We can therefore conclude that the unneeded and unwanted terms that are linear in p^μ can be eliminated simply by taking $\beta^\mu = \gamma^\mu$. So we have now:

$$p^\mu p_\mu - m^2 = (\gamma^\mu p_\mu + m)(\gamma^\omega p_\omega - m)$$

which we will write out explicitly on both sides as:

$$\begin{aligned} (p^0)^2 - (p^1)^2 - (p^2)^2 - (p^3)^2 - m^2 \\ = (\gamma^0)^2 (p^0)^2 + (\gamma^1)^2 (p^1)^2 + (\gamma^2)^2 (p^2)^2 + (\gamma^3)^2 (p^3)^2 - m^2 + \text{CTs} \end{aligned}$$

where we have neglected the cross-terms (CTs) and have used, for example, $(p_1)^2 = (p^1)^2$. We can match each term on LHS with one on the RHS if we arrange that:

$$(\gamma^0)^2 = 1, \quad (\gamma^1)^2 = -1, \quad (\gamma^2)^2 = -1, \quad (\gamma^3)^2 = -1 \quad (1)$$

Then we will have equality as long as the cross-terms are all zero. The cross terms are of form

$$\gamma^0 \gamma^1 p_0 p_1 + \gamma^1 \gamma^0 p_1 p_0$$

And now we have to face a bit of cold reality. The terms p_0 and p_1 are coefficients of a four vector, just numbers, so they commute. If γ^0 and γ^1 are simply coefficients, numbers, than they commute too, and so the cross terms simplify to quantities like:

$$2\gamma^0 \gamma^1 p_0 p_1$$

Now as the cross-terms have to be zero for all possible p^μ , we would need $\gamma^0 \gamma^1 = 0$, which would mean either $\gamma^0 = 0$ for $\gamma^1 = 0$. But this is impossible to reconcile with the condition that $(\gamma^0)^2 = 1$ and $(\gamma^1)^2 = -1$. But what if the γ^μ are matrices? In this case, they need not commute, and the cross terms are of the form:

$$(\gamma^\mu \gamma^\omega + \gamma^\omega \gamma^\mu) p_\mu p_\omega$$

Notice that we have still commuted the p_μ and p_ω which are still the components of four-vectors, i.e. just numbers. So we need matrices with the property that:

$$\gamma^\mu \gamma^\omega + \gamma^\omega \gamma^\mu = 0, \quad (u \neq w) \quad (2)$$

Using the **anti-commutator** between A and B defined as:

$$\{A, B\} \equiv AB + BA$$

we can compactly express the requirements of Equations 1 and 2 as:

$$\{\gamma^\mu, \gamma^\omega\} = \gamma^\mu \gamma^\omega + \gamma^\omega \gamma^\mu = 2g^{\mu\omega} \quad (3)$$

This is the defining characteristic of the gamma group. Any set of objects that satisfy the commutator relationship will do. We will use the Dirac basis (D) in which these are 4x4 matrices of complex numbers:

$$[\gamma^0]_D = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & -1 \end{pmatrix} \quad [\gamma^1]_D = \begin{pmatrix} 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \\ 0 & -1 & 0 & 0 \\ -1 & 0 & 0 & 0 \end{pmatrix}$$

and:

$$[\gamma^2]_D = \begin{pmatrix} 0 & 0 & 0 & -i \\ 0 & 0 & i & 0 \\ 0 & i & 0 & 0 \\ -i & 0 & 0 & 0 \end{pmatrix} \quad [\gamma^3]_D = \begin{pmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & -1 \\ -1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{pmatrix}.$$

There are an infinite number of matrices that satisfy these requirements but the smallest dimensions are 4x4 in dimension, as for the Dirac basis. When the choice of Dirac basis is clear, we will often omit the $[\dots]_D$, which is more conventional presentation but less explicit.

There are more compact ways to write the gamma matrices, while still remaining in the Dirac basis.

$$[\gamma^0]_D = \begin{pmatrix} I & 0 \\ 0 & -I \end{pmatrix}, \quad [\gamma^i]_D = \begin{pmatrix} 0 & \sigma^i \\ -\sigma^i & 0 \end{pmatrix} \quad (i = 1, 2, 3) \quad (4)$$

where I is the 2x2 identity matrix:

$$I = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

and the three σ^i are the Pauli spin matrices:

$$\sigma^1 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad \sigma^2 = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \quad \sigma^3 = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

With a solution in hand, we now return to our original problem, which was to factor the relativistic energy equation as:

$$p^\mu p_\mu - m^2 = (\gamma^\mu p_\mu - m)(\gamma^\omega p_\omega + m) = 0$$

Setting either of the two factors equal to zero will set the entire equation to zero (as needed). It only makes a superficial difference which one we choose, but the conventional choice is to set:

$$\gamma^\mu p_\mu - m = 0$$

Now we can make the substitution:

$$p_\mu \rightarrow i\hbar \partial_\mu$$

To obtain the Dirac Equation:

$$i\hbar \gamma^\mu \partial_\mu \psi - m\psi = 0 \quad (5)$$

Notice that ψ can not be a scalar function in the Dirac basis, for that would have us adding a scalar to a 4x4 matrix. The simplest solution is for ψ to be a four element column matrix:

$$[\psi]_D = \begin{pmatrix} \psi_1 \\ \psi_2 \\ \psi_3 \\ \psi_4 \end{pmatrix}$$

Despite having four components, ψ is **not** a four-vector because (as we will show later) it does not transform like a four-vector.

5 Solutions to the Dirac Equation

We'll find solutions to the Dirac equation:

$$i\hbar\gamma^\mu\partial_\mu\psi - m\psi = 0$$

Let's first consider solutions with $\vec{p} = 0$, so:

$$\frac{\partial\psi}{\partial x} = \frac{\partial\psi}{\partial y} = \frac{\partial\psi}{\partial z} = 0$$

and the Dirac equation becomes:

$$\gamma^0\frac{\partial\psi}{\partial t} = -i\frac{m}{\hbar}\psi$$

Writing this out the long way in the Dirac representation:

$$\begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & -1 \end{pmatrix} \frac{\partial}{\partial t} \begin{pmatrix} \psi_1 \\ \psi_2 \\ \psi_3 \\ \psi_4 \end{pmatrix} = -i\frac{m}{\hbar} \begin{pmatrix} \psi_1 \\ \psi_2 \\ \psi_3 \\ \psi_4 \end{pmatrix}$$

or

$$\frac{\partial}{\partial t} \begin{pmatrix} \psi_1 \\ \psi_2 \\ -\psi_3 \\ -\psi_4 \end{pmatrix} = -i\frac{m}{\hbar} \begin{pmatrix} \psi_1 \\ \psi_2 \\ \psi_3 \\ \psi_4 \end{pmatrix}$$

There are four independent solutions:

$$[\psi^{(1)}]_D = N \exp\left(-i\frac{m}{\hbar}t\right) \begin{pmatrix} 1 \\ 0 \\ 0 \\ 0 \end{pmatrix}, \quad [\psi^{(2)}]_D = N \exp\left(-i\frac{m}{\hbar}t\right) \begin{pmatrix} 0 \\ 1 \\ 0 \\ 0 \end{pmatrix},$$

and

$$[\psi^{(3)}]_D = N \exp\left(+i\frac{m}{\hbar}t\right) \begin{pmatrix} 0 \\ 0 \\ 1 \\ 0 \end{pmatrix}, \quad [\psi^{(4)}]_D = N \exp\left(+i\frac{m}{\hbar}t\right) \begin{pmatrix} 0 \\ 0 \\ 0 \\ 1 \end{pmatrix},$$

where N is a normalization factor.

Recalling that the energy operator is:

$$E = i\hbar \frac{\partial}{\partial t}$$

We see that:

$$i\hbar \frac{\partial \Psi^{(1)}}{\partial t} = m\Psi^{(1)} \quad i\hbar \frac{\partial \Psi^{(2)}}{\partial t} = m\Psi^{(2)}$$

and:

$$i\hbar \frac{\partial \Psi^{(3)}}{\partial t} = -m\Psi^{(3)} \quad i\hbar \frac{\partial \Psi^{(4)}}{\partial t} = -m\Psi^{(4)}$$

So evidently $\Psi^{(1)}$ and $\Psi^{(2)}$ have energy $E = m$ while $\Psi^{(3)}$ and $\Psi^{(4)}$ have energy $E = -m$.

I like to imagine Dirac laughing to himself at this point. For despite the great lengths he went through to develop a wave function that is first order in Energy, we still arrive at negative energy solutions. And we cannot simply throw these solutions out on the grounds that they are non-physical, because we would no longer have a complete basis for our Hilbert space of bispinors. As we'll see soon, these negative energy solutions imply the existence antiparticles. However, even setting aside the negative energy solutions for now, that still leaves two solutions. As we'll see later, this implies that the Dirac equation applies to spin one-half particles (like the electron), and the two positive energy solutions that we obtained are for spin up and spin down states.

We should note that these solutions do not have the problem we so readily encountered with solutions to the K-G equation. Specifically, for our solutions to a particle of mass m at rest we see that: $|\Psi|^2 = |N|^2 > 0$ independent of time for all four of these solutions. More generally, it can be shown that

$$\frac{d}{dt} \int |\Psi(x, t)|^2 d^3x = 0$$

for all solutions to the Dirac equation, just as for the non-relativistic Schrodinger equation, and $|\Psi(x, t)|^2 \geq 0$.

6 Antiparticles

Dirac postulated that the negative energy solutions represented actual physical states. To reconcile the existence of these states with physical reality, he postulated that in the vacuum, the negative energy states are all occupied, as shown in Fig. 1. The Fermi exclusion principle then prevents ordinary particles from falling down into the negative energy states, releasing an unlimited amount of energy.

In this scenario, adding sufficient energy to a particle in a negative energy state, will raise it to a positive energy state, and so create an ordinary particle. In so doing, it will leave behind a vacancy in the Dirac sea. The vacancy would behave as if a particle with opposite charge to the ordinary particle had been created ($0 = q + (-q)$). This is a very workable (if implausible) description for the creation of particle antiparticle pairs, and indeed it was sufficient to predict the existence and properties of the positron, which was later discovered by Anderson.

The energy eigenstates of particles with $E > 0$ have time-dependence:

$$\exp(-iEt/\hbar) = \exp(-i|E|t/\hbar)$$

whereas particles with $E < 0$ have time-dependence:

$$\exp(-iEt/\hbar) = \exp(+i|E|t/\hbar)$$

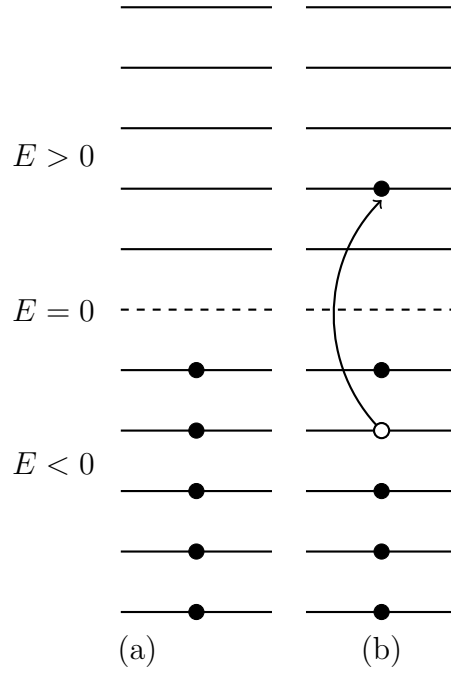


Figure 1: In the vacuum (a) the negative energy states are all filled, forming the Dirac Sea. A particle in a negative energy state can gain energy, occupying a positive energy state to produce a particle, and leaving behind a vacancy (or hole) in the Dirac sea, which is interpreted as an antiparticle.

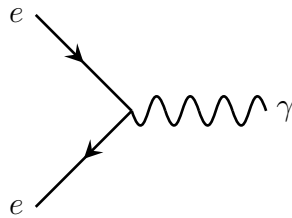


Figure 2: This Feynman diagram has three equivalent interpretations: (1) two incident electrons, one with positive and one with negative energy, (2) two incident electrons, one propagating forward in time, and one propagating backward, (3) an incident electron and positron.

This suggests that particles with negative energy can be reinterpreted as particles with positive energy that propagate backwards in time. We derived the Dirac equation to be consistent with special relativity, and we have already seen that time-reversal is a member of the Lorentz group.

This is best visualized in the context of a Feynman diagram, as in Fig. 2. In the Feynman-Stueckelberg interpretation, particle states propagating backwards in time are interpreted as antiparticle states propagating forward in time. Whereas an incident particle of charge q propagating forward in time brings a charge q , an incident particle of charge q propagating backward in time carries away a charge q . Thus we expect antiparticles to carry the opposite charge of the corresponding particle. The energy of both particle and antiparticle is positive, so the mass of the antiparticle does not change sign, and we expect both particle and antiparticle to have the same positive mass.

Particles wave functions have time-dependence:

$$\exp(-iEt/\hbar)$$

and antiparticles wave functions have time dependence:

$$\exp(iEt/\hbar)$$

where $E > 0$ in both cases is the physical energy of the particle. Particles have momentum operator given by:

$$p_\mu = i\hbar\partial_\mu$$

which leads to:

$$p^0 = E$$

as expected. Antiparticles must therefore have:

$$p_\mu = -i\hbar\partial_\mu$$

which leads to:

$$p^0 = E$$

as expected. The angular momentum operator, through its dependence on momentum, will also change sign. Spin also changes sign, otherwise, we could distinguish a particle with intrinsic spin one from a composite system with total angular-momentum one.

7 Notation

There is some new notation to help us navigate the strange new world Dirac revealed by taking his square-root. Because the three Pauli matrices ($\sigma^1, \sigma^2, \sigma^3$) are often associated with the three spatial coordinates (x, y, z) we can consider them as forming a sort of three-vector:

$$\vec{\sigma} \equiv \sigma^1 \hat{x} + \sigma^2 \hat{y} + \sigma^3 \hat{z}$$

whose components are matrices instead of scalars. This three-vector can be dotted with other three vectors, e.g.:

$$\vec{a} \cdot \vec{\sigma} = a_x \sigma^1 + a_y \sigma^2 + a_z \sigma^3 = \begin{pmatrix} a_z & a_x - ia_y \\ a_x + ia_y & -a_z \end{pmatrix}$$

It is useful to note that:

$$\begin{aligned} (\vec{a} \cdot \vec{\sigma})^2 &= (a_x \sigma^1 + a_y \sigma^2 + a_z \sigma^3)^2 \\ &= a_x^2 (\sigma^1)^2 + a_y^2 (\sigma^2)^2 + a_z^2 (\sigma^3)^2 + \text{CTs} \end{aligned}$$

The cross-terms (CTs) are all zero, as the Pauli matrices anti-commute, for example:

$$a_x a_y (\sigma^1 \sigma^2 + \sigma^2 \sigma^1) = 0$$

Recalling that also:

$$(\sigma^1)^2 = (\sigma^2)^2 = (\sigma^3)^2 = 1$$

we conclude:

$$(\vec{a} \cdot \vec{\sigma})^2 = |\vec{a}|^2$$

We encounter gamma matrices contracted with four vectors so often the combinations gets its own notation, the Feynman slash notation:

$$\begin{aligned} \not{a} &\equiv \gamma^\mu a_\mu = a^0 \begin{pmatrix} I & 0 \\ 0 & -I \end{pmatrix} - \vec{a} \cdot \begin{pmatrix} 0 & \vec{\sigma} \\ -\vec{\sigma} & 0 \end{pmatrix} \\ &= \begin{pmatrix} a^0 I & -\vec{a} \cdot \vec{\sigma} \\ \vec{a} \cdot \vec{\sigma} & -a^0 I \end{pmatrix} \end{aligned}$$

While we are discussing notation, we won't be writing out I explicitly much longer, when it can be inferred from context. The above, for example, would usually be written as:

$$\begin{aligned} \not{a} &\equiv \gamma^\mu a_\mu = a^0 \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} - \vec{a} \cdot \begin{pmatrix} 0 & \vec{\sigma} \\ -\vec{\sigma} & 0 \end{pmatrix} \\ &= \begin{pmatrix} a^0 & -\vec{a} \cdot \vec{\sigma} \\ \vec{a} \cdot \vec{\sigma} & -a^0 \end{pmatrix} \end{aligned}$$

8 Plane Wave Solutions for Particles

We are now ready to tackle the plane wave solutions. You may recall from NRQM that these are all we will need for describing scattering experiments, since our experiments measure particles only far away from the interaction potentials, at infinity effectively, where they are free particles well described by plane waves (or a wave packet from the superposition of plane waves).

We will therefore attempt to find particle solutions of the form:

$$\psi = a e^{-ikx} u(k)$$

where a is a normalization factor, and the exponential contains a factor

$$kx = k^\mu x_\mu$$

which is the only space and time dependence. The $u(k)$ is the bi-spinor part which we can write as either two spinors or four components:

$$[u(k)]_D \equiv \begin{pmatrix} u_A \\ u_B \end{pmatrix} \equiv \begin{pmatrix} u_1 \\ u_2 \\ u_3 \\ u_4 \end{pmatrix}$$

The components are still dependent on k but we have dropped the explicit dependence for brevity.

Note that:

$$\partial_\mu \psi = \frac{\partial \psi}{\partial x^\mu} = a \left(\frac{\partial}{\partial x^\mu} \exp(-ik_\mu x^\mu) \right) u(k) = -ik_\mu \psi$$

The Dirac equation:

$$i\hbar\gamma^\mu\partial_\mu\psi - m\psi = 0$$

when applied to the plane wave yields:

$$\begin{aligned}(\hbar\gamma^\mu k_\mu - m)\psi &= 0 \\(\hbar\gamma^\mu k_\mu - m)a e^{-ikx}u(k) &= 0 \\(\hbar\not{k} - m)u(k) &= 0 \\ \begin{pmatrix} \hbar k^0 - m & -\hbar\vec{k} \cdot \vec{\sigma} \\ \hbar\vec{k} \cdot \vec{\sigma} & -\hbar k^0 - m \end{pmatrix} \begin{pmatrix} u_A \\ u_B \end{pmatrix} &= 0\end{aligned}$$

Solving the last matrix equation we obtain:

$$u_A = \frac{\hbar\vec{k} \cdot \vec{\sigma}}{\hbar k^0 - m} u_B \quad (6)$$

$$u_B = \frac{\hbar\vec{k} \cdot \vec{\sigma}}{\hbar k^0 + m} u_A \quad (7)$$

Plugging the second equation into the first:

$$\begin{aligned}u_A &= \frac{(\hbar\vec{k} \cdot \vec{\sigma})^2}{(\hbar k^0)^2 - (m)^2} u_A \\ &= \frac{\hbar^2 |\vec{k}|^2}{(\hbar k^0)^2 - (m)^2} u_A \\ (m)^2 &= \hbar^2 ((k^0)^2 - |\vec{k}|^2) = \hbar^2 k^\mu k_\mu\end{aligned}$$

But we already know a four-vector whose square is m^2 , the four-momentum:

$$p^2 = (m)^2$$

We conclude that $p^\mu = \pm \hbar k^\mu$. However, only $p^\mu = \hbar k^\mu$ leads to the correct time-dependence

$$\psi = a e^{-ikx} u(k) = a \exp(-iEt/\hbar) \exp(i\vec{p} \cdot \vec{x}) u(k)$$

for a particle of energy E and momentum $\vec{p}$. We may rewrite Equation 7 equivalently as:

$$u_B = \frac{\vec{\sigma} \cdot \vec{p}}{E + m} u_A \quad (8)$$

and writing out $\vec{\sigma} \cdot \vec{p}$ as a matrix:

$$u_B = \frac{1}{E + m} \begin{pmatrix} p_z & p_x - ip_y \\ p_x + ip_y & -p_z \end{pmatrix} u_A$$

And next we find two solutions by applying this result to a starting choice for u_A :

$$\begin{aligned}u_A = \begin{pmatrix} 1 \\ 0 \end{pmatrix} &\implies u_B = \frac{1}{E + m} \begin{pmatrix} p_z \\ p_x + ip_y \end{pmatrix} \\ u_A = \begin{pmatrix} 0 \\ 1 \end{pmatrix} &\implies u_B = \frac{1}{E + m} \begin{pmatrix} p_x - ip_y \\ -p_z \end{pmatrix}\end{aligned}$$

For particles, we can go no further. If we attempt to make a choice for u_B , then we would be obliged to use Equation 6, which will blow up when $E = m$ due to the negative sign in the denominator.

We assemble u_A and u_B to form the entire bi-spinor for the particle solutions:

$$[u^{(1)}]_D = N \begin{pmatrix} 1 \\ 0 \\ \frac{p_z}{E+m} \\ \frac{p_x + ip_y}{E+m} \end{pmatrix}, \quad [u^{(2)}]_D = N \begin{pmatrix} 0 \\ 1 \\ \frac{p_x - icp_y}{E+m} \\ \frac{-p_z}{E+m} \end{pmatrix}.$$

where N is a not-yet-determined normalization factor.

9 Plane Wave Solutions for Antiparticles

Next we attempt to find antiparticle solutions of the form:

$$\psi = a e^{ikx} v(k)$$

where a is a normalization factor, and the exponential contains a factor

$$kx = k^\mu x_\mu$$

which is the only space and time dependence. The $v(k)$ is the bi-spinor part which we can write as either two spinors or four components:

$$[v(k)]_D \equiv \begin{pmatrix} v_A \\ v_B \end{pmatrix} \equiv \begin{pmatrix} v_1 \\ v_2 \\ v_3 \\ v_4 \end{pmatrix}$$

In this case the Dirac equation

$$i\hbar\gamma^\mu\partial_\mu\psi - m\psi = 0$$

when applied to the plane wave yields:

$$\begin{aligned} (-\hbar\gamma^\mu k_\mu - m) a e^{ikx} v(k) &= 0 \\ (\hbar\cancel{k} + m) v(k) &= 0 \\ \begin{pmatrix} \hbar k^0 + m & -\hbar\vec{k} \cdot \vec{\sigma} \\ \hbar\vec{k} \cdot \vec{\sigma} & -\hbar k^0 + m \end{pmatrix} \begin{pmatrix} v_A \\ v_B \end{pmatrix} &= 0 \end{aligned}$$

Solving the first row of matrix equation we obtain:

$$v_A = \frac{\hbar\vec{k} \cdot \vec{\sigma}}{\hbar k^0 + m} v_B \quad (9)$$

By a similar calculation as in the previous section we conclude that: $p^\mu = \pm\hbar k^\mu$. However, only $p^\mu = \hbar k^\mu$ leads to the correct time-dependence

$$\psi = a e^{ikx} v(k) = a \exp(iEt/\hbar) \exp(-i\vec{p} \cdot \vec{x}) v(k)$$

for an antiparticle of energy E and momentum $\vec{p}$. We may rewrite Equation 9 as:

$$v_A = \frac{\vec{\sigma} \cdot \vec{p}}{p^0 + m} v_B = \frac{1}{E + m} \vec{\sigma} \cdot \vec{p} v_B$$

and writing out $\vec{\sigma} \cdot \vec{p}$ as a matrix:

$$v_A = \frac{1}{E + m} \begin{pmatrix} p_z & p_x - ip_y \\ p_x + ip_y & -p_z \end{pmatrix} v_B$$

And next we find two solutions by applying this result to a starting choice for v_B :

$$\begin{aligned} v_B = \begin{pmatrix} 1 \\ 0 \end{pmatrix} &\implies v_A = \frac{1}{E + m} \begin{pmatrix} p_z \\ p_x + ip_y \end{pmatrix} \\ v_B = \begin{pmatrix} 0 \\ 1 \end{pmatrix} &\implies v_A = \frac{1}{E + m} \begin{pmatrix} p_x - ip_y \\ -p_z \end{pmatrix} \end{aligned}$$

We assemble v_A and v_B to form the entire bi-spinor for the antiparticle solutions:

$$[v^{(1)}]_D = N \begin{pmatrix} \frac{p_x - ip_y}{E + m} \\ \frac{-p_z}{E + m} \\ 0 \\ 1 \end{pmatrix}, \quad [v^{(2)}]_D = -N \begin{pmatrix} \frac{p_z}{E + m} \\ \frac{p_x + ip_y}{E + m} \\ 1 \\ 0 \end{pmatrix},$$

The rationale for ordering the solutions this way, and the reason for the overall negative sign in front of $v^{(2)}$ will become clear in the exercises (e.g. Griffiths P7.9).

We will use u for particles and v for antiparticles. The Dirac equation for particles in terms of the momentum now reads:

$$(\not{p} - m)u = 0$$

For antiparticles, $k^\mu = -p^\mu/\hbar$, and the Dirac equation reads:

$$(\not{p} + m)v = 0$$

10 Review: Angular Momentum in Quantum Mechanics

This section follows Griffith's Introduction to Quantum Mechanics, Section 4.3, so you should refer to that if you get stuck on any of the exercises.

The classical angular momentum is:

$$\vec{L} = \vec{r} \times \vec{p}$$

from which we determine the corresponding angular momentum operators using the position and momentum operators:

$$L_x \equiv yp_z - zp_y, \quad L_y \equiv zp_x - xp_z, \quad L_z \equiv xp_y - yp_x$$

and the total momentum squared:

$$L^2 = L_x^2 + L_y^2 + L_z^2$$

In the exercises, you will verify that:

$$[L_x, L_y] = i\hbar L_z, \quad [L_y, L_z] = i\hbar L_x, \quad [L_z, L_x] = i\hbar L_y \quad (10)$$

and

$$[L^2, L_x] = [L^2, L_y] = [L^2, L_z] = 0 \quad (11)$$

This shows that while L_x , L_y , and L_z are incompatible observables, L^2 commutes with any single component. So we can therefore find eigenstates of both L^2 and L_z .

Equation 10 reveals that there is a closed algebra of commutators over the angular momentum operators. We use an algebraic approach based on the ladder operators:

$$L_{\pm} \equiv L_x \pm iL_y$$

for which you will show:

$$[L^2, L_{\pm}] = 0 \quad (12)$$

and:

$$[L^2, L_{\pm}] = \pm \hbar L_{\pm} \quad (13)$$

and:

$$L^2 = L_{\pm}L_{\mp} - L_z^2 \mp \hbar L_z \quad (14)$$

Exercise 9: Prove Equations 10. You need only prove it for one, then the rest follow from the cyclic rotation $xyz \rightarrow yzx$ which leaves the definitions unchanged.

Exercise 10: Prove Equations 11. You need only prove it for one, then the rest follow from the cyclic rotation $xyz \rightarrow yzx$.

Exercise 11: Prove Equations 12

Exercise 12: Prove Equation 13

Exercise 13: Prove Equation 14

With the heavy lifting behind us, we can consider a simultaneous eigenstate of both L^2 and L_z so that:

$$L^2 f = \alpha f$$

and

$$L_z f = \mu f$$

Consider the state $L_+ f$. Using Equation 12 we see that:

$$L^2(L_+ f) = L_+(L^2 f) = L_+(\alpha f) = \alpha(L_+ f)$$

and using Equation 13 we see that:

$$L_z(L_+ f) = (L_z L_+ + \hbar L_+) f = (\mu + \hbar)(L_+ f)$$

This shows that $L_+ f$ remains an eigenstate of L^2 with the same eigenvalue as f , and also remains an eigenstate of L_z but with new eigenvalue $\mu + \hbar$. We say that L_+ is a raising operator.

We cannot keep applying the raising operator indefinitely, for eventually the square of the z component of angular momentum would be larger than the L^2 . Therefore, after some number of applications of L_+ we must eventually reach a state g for which:

$$L_+ g = 0$$

For convenience in what follows, let's define a dimensionless quantity ℓ by the eigenvalue of this “top-rung” state g :

$$L_+g \equiv \ell\hbar g$$

In the exercises, you will show that:

$$L^2g = \hbar^2\ell(\ell+1)g$$

which relates the eigenvalue of L^2 (which we do not change via the raising operator) to the maximum eigenvalue of the L_z .

In a similar fashion, we see that the operator L_- is a lowering operator, so that L_-f remains an eigenstate of L^2 with same eigenvalue, and also remains an eigenstate of L_z but with new eigenvalue $\mu - \hbar$. Also similarly, we cannot endlessly apply L_- . After some number of applications of L_- we must reach a state h for which:

$$L_-h = 0$$

Again for convenience, we define a dimensionless quantity $\bar{\ell}$ by the eigenvalue of the “bottom-rung” state h :

$$L_zh \equiv \bar{\ell}\hbar h$$

In the exercises you will show that

$$L^2h = \hbar^2\bar{\ell}(\bar{\ell}-1)h$$

Exercise 14: Using equation 14 show that:

$$L^2g = \hbar^2\ell(\ell+1)g$$

and

$$L^2h = \hbar^2\bar{\ell}(\bar{\ell}-1)h$$

□

Comparing these results for the lower-rung with those of the upper rung, and noting that the raising operator leaves the the eigenvalue of L^2 unchanged, we conclude that $\ell(\ell+1) = \bar{\ell}(\bar{\ell}-1)$. So either

$$\bar{\ell} = \ell + 1$$

which is impossible, since we must have the lower bound below the upper bound:

$$\bar{\ell}\hbar < \ell\hbar$$

or

$$\bar{\ell} = -\ell$$

This implies that ℓ is non-negative. Furthermore, we move from $-\ell$ to ℓ in an integer number of applications of N , so:

$$-\ell + N = \ell$$

which implies that ℓ is integer or half-integer:

$$\ell = \frac{N}{2}$$

To summarize, the eigenstates $|lm\rangle$ of L^2 are L_z are characterized by two quantum numbers:

$$\ell = 0, \frac{1}{2}, 1, \frac{3}{2}, 2, \dots$$

and

$$m = -\ell, -\ell + 1, \dots, \ell$$

with:

$$L^2 |lm\rangle = \hbar^2 \ell(\ell + 1) |lm\rangle$$

and

$$L_z |lm\rangle = \hbar m |lm\rangle$$

In the last exercise of this section, you will show that:

$$L_+ |lm\rangle = \hbar \sqrt{(l-m)(l+m+1)} |l(m+1)\rangle$$

Similarly:

$$L_- |lm\rangle = \hbar \sqrt{(l+m)(l-m+1)} |l(m-1)\rangle$$

Exercise 15: Show that:

$$L_+ |lm\rangle = \hbar \sqrt{(l-m)(l+m+1)} |l(m+1)\rangle$$

Hint: First put:

$$L_+ |lm\rangle = \alpha |lm\rangle$$

where α is the quantity we seek to calculate. Notice that $L_+^\dagger = L_-$ and so:

$$\langle lm | L_- L_+ | lm \rangle = \langle lm | L_+^\dagger L_+ | lm \rangle = |\alpha|^2$$

Then use Equation 14. $\square$

Although we will not do so in this class, if you find the Energy eigenstates for a central potential, such as the Hydrogen atom, using separation of variables for the non-relativistic Schrodinger equation, you only find integer values of ℓ :

$$\ell = 0, 1, 2, \dots$$

And yet you find a complete set of energy eigenstates. Evidently orbital angular momentum from the motion of particles in 3-dimensional space only admits integer values of ℓ .

However, as the Dirac equation will eventually reveal to us, nature does make use of the half-odd integer values for angular momentum as well. Fundamental particles have an intrinsic angular momentum called spin. The electron, for example, has spin one-half. For spin, we'll use quantum number:

$$s = 0, \frac{1}{2}, 1, \frac{3}{2}, 2, \dots$$

and

$$m = -s, (-s + 1), \dots, s$$

11 Spin One-Half

For a spin one-half system, there are two states which are simultaneous eigenstates of the S^2 and S_z operators. The spin up state has $s = 1/2$ and $m = 1/2$ which we write as:

$$|\uparrow\rangle = \left| \frac{1}{2}, \frac{1}{2} \right\rangle$$

and the spin down state has $s = 1/2$ and $m = -1/2$ which we write as:

$$|\downarrow\rangle = |\frac{1}{2}, -\frac{1}{2}\rangle$$

They have the defining properties:

$$S_z |\uparrow\rangle = \frac{\hbar}{2} |\uparrow\rangle$$

and

$$S_z |\downarrow\rangle = -\frac{\hbar}{2} |\downarrow\rangle$$

These two states are a basis for the spin one-half system, which we will call the S_z basis. We could of course form equivalent basis from the eigenstates of the S_x operator, for example. But we can write any state (even, for example, an eigenstate of S_x) as a linear combination of these two basis vectors. As always, given a basis, we can represent a vector as a column vector in that basis. So we have:

$$[|\uparrow\rangle]_{S_z} = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$$

and

$$[|\downarrow\rangle]_{S_z} = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

The linear operator S_z can be represented as a matrix in this base:

$$[S_z]_{S_z} = \frac{\hbar}{2} \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} = \frac{\hbar}{2} \sigma^3$$

The raising and lowering operators are:

$$S_{\pm} = S_x \pm iS_y$$

and we have:

$$S_+ |\downarrow\rangle = \hbar |\uparrow\rangle \quad S_+ |\uparrow\rangle = 0$$

and:

$$S_- |\uparrow\rangle = \hbar |\downarrow\rangle \quad S_- |\downarrow\rangle = 0$$

from which we infer:

$$[S_+]_{S_z} = \hbar \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$$

and

$$[S_-]_{S_z} = \hbar \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}.$$

We also have:

$$S_x = \frac{S_+ + S_-}{2}$$

and

$$S_y = \frac{S_+ - S_-}{2i}$$

and so:

$$[S_x]_{S_z} = \frac{\hbar}{2} \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} = \frac{\hbar}{2} \sigma^1$$

and

$$[S_y]_{S_z} = \frac{\hbar}{2} \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} = \frac{\hbar}{2} \sigma^2$$

How can we extend our spin-1/2 operator to our four-component bispinors? Suppose we have a group G with $a, b, c \in G$ and $ab = c$. We can always build a new group by taking putting each element of G along the diagonal of a 2x2 matrix, for example:

$$a \rightarrow \begin{pmatrix} a & 0 \\ 0 & a \end{pmatrix}, \quad b \rightarrow \begin{pmatrix} b & 0 \\ 0 & b \end{pmatrix}, \quad c \rightarrow \begin{pmatrix} c & 0 \\ 0 & c \end{pmatrix},$$

the new group will preserve all the features of the old group because:

$$\begin{pmatrix} a & 0 \\ 0 & a \end{pmatrix} \begin{pmatrix} b & 0 \\ 0 & b \end{pmatrix} = \begin{pmatrix} ab & 0 \\ 0 & ab \end{pmatrix} = \begin{pmatrix} c & 0 \\ 0 & c \end{pmatrix}$$

We can therefore construct the S_z operator appropriate for a bi-spinor by recycling the definition appropriate for a spinor:

$$S_z = \frac{\hbar}{2} \begin{pmatrix} \sigma^3 & 0 \\ 0 & \sigma^3 \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & -1 \end{pmatrix}$$

With this operator in hand, we see that $\psi^{(1)}$ has spin up, and $\psi^{(2)}$ has spin down. For the antiparticle solutions, it turns out the appropriate operator is $-\hat{S}_z$, and so $\psi^{(3)}$ has spin down, and $\psi^{(4)}$ has spin up.

Exercise 16: (*) The spin-squared operator is:

$$S^2 = S_x^2 + S_y^2 + S_z^2$$

Calculate

$$S^2 |\uparrow\rangle$$

and

$$S^2 |\downarrow\rangle$$

explicitly in the Dirac basis. $\square$

Exercise 17: (*) Verify that:

$$[S_x, S_y] = i\hbar S_z$$

explicitly in the Dirac basis. $\square$

Exercise 18: (*) Verify that:

$$S_+ |\downarrow\rangle = \hbar |\uparrow\rangle$$

and

$$S_+ |\uparrow\rangle = 0$$

explicitly in the Dirac basis. $\square$

12 Conservation of Total Angular Momentum in the Dirac Equation

In this section we'll show conclusively that the Dirac equation applies to spin one-half particles. We'll proceed as follows. First, we'll find the Hamiltonian operator corresponding to the Dirac equation. Then we'll show that orbital angular momentum L does not commute with H , and so it is not a compatible observable. Last, we'll show that $L + S$, the orbital angular momentum plus an intrinsic angular momentum of spin one-half does commute with H .

Starting from:

$$i\hbar\gamma^\mu\partial_\mu\psi - m\psi = 0$$

we have

$$\begin{aligned} i\hbar\gamma^0\frac{\partial}{\partial t} &= i\hbar\gamma^i\frac{\partial}{\partial x^i} - m \\ &= i\hbar\gamma^ip_i - m \end{aligned}$$

and so in the Dirac representation:

$$[H]_D = \begin{pmatrix} m & \vec{\sigma} \cdot \vec{p} \\ \vec{\sigma} \cdot \vec{p} & -m \end{pmatrix}$$

but we will drop the $[\dots]_D$ in what follows. Using:

$$L_z = xp_y - yp_x$$

First finding the commutator with each component of H we have:

$$[m, L_z] = 0$$

then

$$\begin{aligned} [\vec{\sigma} \cdot \vec{p}, L_z] &= [\sigma^x p_x + \sigma^y p_y, xp_y - yp_x] \\ &= -i\hbar\sigma^x p_y - \sigma^y p_x \\ &= -i\hbar[\vec{\sigma} \times \vec{p}]_z \end{aligned}$$

So then:

$$[H, L_z] = -i\hbar \begin{pmatrix} 0 & [\vec{\sigma} \times \vec{p}]_z \\ [\vec{\sigma} \times \vec{p}]_z & 0 \end{pmatrix}$$

Now using:

$$S_z = \frac{\hbar}{2} \begin{pmatrix} \sigma^z & 0 \\ 0 & \sigma^z \end{pmatrix}$$

and finding the component-wise commutators first:

$$\begin{aligned} [\vec{\sigma} \cdot \vec{p}, \sigma^z] &= [\sigma^x p_x + \sigma^y p_y, \sigma^z] \\ &= 2i(-\sigma^y p_x + \sigma^x p_y) \\ &= 2i[\vec{\sigma} \times \vec{p}]_z \end{aligned}$$

where we have used cyclic rotations of:

$$[\sigma^x \sigma^y] = 2i\sigma^z$$

and so:

$$[H, S_z] = i\hbar \begin{pmatrix} 0 & [\vec{\sigma} \times \vec{p}]_z \\ [\vec{\sigma} \times \vec{p}]_z & 0 \end{pmatrix}$$

At last we conclude:

$$[H, L_z + S_z] = 0$$

Similarly:

$$[H, L_x + S_x] = [H, L_y + S_y] = 0$$

Exercise 19: (*) Show explicitly that the particle solutions $u^{(1)}$ and $u^{(2)}$ are eigenvectors of the matrix representation for the Hamiltonian. $\square$

Exercise 20: For a particle at rest ($\vec{p} = 0$) categorize $u^{(1)}$, $u^{(2)}$, $v^{(1)}$, $v^{(2)}$ as either spin-up or spin down.

13 Relativistic Transformation of Bi-Spinors

Suppose that two frames are related by a Lorentz transformation:

$$x^{\mu'} = \Lambda^{\mu'}_{\nu} x^{\nu}$$

In this section, we are going to determine the properties of the transformation S which relates the bispinors appropriate to each frame:

$$\psi' = S\psi$$

and S must be invertible so that we have:

$$\psi = S^{-1}\psi'$$

The Dirac equation, which we derived to be consistent with special relativity, must be equivalent in all reference frames:

$$i\hbar\gamma^{\mu}\partial_{\mu}\psi = m\psi \Leftrightarrow i\hbar\gamma^{\beta'}\partial_{\beta'}\psi' = m\psi'$$

Notice that we cannot use our primed indices notation on ψ , because ψ does not have any indices, but yet it is not a scalar. It is a bispinor, with four components, but it is not a four vector. Hence we use ψ' to refer to the bispinor with components appropriate to the frame F' .

Since the γ and δ tensors are invariant, we have:

$$\gamma^{\beta'} = \delta^{\beta'}_{\alpha} \gamma^{\alpha}$$

and also:

$$\begin{aligned} \partial_{\beta'} &= \Lambda^{\mu}_{\beta'} \partial_{\mu} \\ \psi' &= S\psi \end{aligned}$$

So inserting these transformations into the Dirac equation in the primed frame:

$$\begin{aligned}
i\hbar\gamma^{\beta'}\partial_{\beta'}\psi' &= m\psi' \\
i\hbar\delta_{\alpha}^{\beta'}\gamma^{\alpha}\Lambda_{\beta'}^{\mu}\partial_{\mu}S\psi &= mS\psi \\
S^{-1}i\hbar\delta_{\alpha}^{\beta'}\gamma^{\alpha}\Lambda_{\beta'}^{\mu}\partial_{\mu}S\psi &= m\psi \\
i\hbar\delta_{\alpha}^{\beta'}\Lambda_{\beta'}^{\mu}S^{-1}\gamma^{\alpha}S\partial_{\mu}\psi &= i\hbar\gamma^{\mu}\partial_{\mu}\psi
\end{aligned}$$

where we have taken care in the last line to maintain the order of matrices but have moved numbers around. We see that it is sufficient that:

$$\delta_{\alpha}^{\beta'}\Lambda_{\beta'}^{\mu}S^{-1}\gamma^{\alpha}S = \gamma^{\mu}$$

Multiplying each side by $\Lambda_{\mu}^{\omega'}$:

$$\begin{aligned}
(\delta_{\alpha}^{\beta'}\Lambda_{\beta'}^{\mu}\Lambda_{\mu}^{\omega'})S^{-1}\gamma^{\alpha}S &= \Lambda_{\mu}^{\omega'}\gamma^{\mu} \\
\delta_{\alpha}^{\omega'}S^{-1}\gamma^{\alpha}S &= \Lambda_{\mu}^{\omega'}\gamma^{\mu}
\end{aligned}$$

The δ tensor on the LHS is an artifact of our notation which primes indices.

In the exercises, you will work with an explicit transformation matrix S defined as:

$$S \equiv a_{+} + a_{-}\gamma^0\gamma^1$$

where:

$$a_{\pm} = \pm\sqrt{\frac{\gamma \pm 1}{2}}$$

with γ the usual factor for a Lorentz transformation. Remembering to insert identity matrices as needed, we can write S as:

$$S = \begin{pmatrix} a_{+} & a_{-}\sigma^1 \\ a_{-}\sigma^1 & a_{+} \end{pmatrix} = \begin{pmatrix} a_{+} & 0 & 0 & a_{-} \\ 0 & a_{+} & a_{-} & 0 \\ 0 & a_{-} & a_{+} & 0 \\ a_{-} & 0 & 0 & a_{+} \end{pmatrix}$$

which is rather strikingly beautiful, if nothing else!

Exercise 21: (*) Show that:

$$\begin{aligned}
a_{+}^2 - a_{-}^2 &= 1 \\
a_{+}^2 + a_{-}^2 &= \gamma \\
2a_{+}a_{-} &= -|\beta|\gamma
\end{aligned}$$

from which we can see that we can construct every element of the a Lorentz Boost from a_{+} and a_{-} . $\square$

Since σ^1 and I commute, we can calculate the determinant of S as if it were a 2x2 matrix. First calculate:

$$\begin{aligned}\det(S) &= \det((a_+I)^2 - (a_-\sigma^1)^2) \\ &= \det((a_+)(I)^2 - (a_-)^2(\sigma^1)^2) \\ &= \det((a_+^2 - a_-^2)I) \\ &= (a_+^2 - a_-^2) \det(I) \\ &= 1\end{aligned}$$

and so (again working as if S were a 2x2 matrix) we have:

$$S^{-1} = \begin{pmatrix} a_+ & -a_- \sigma^1 \\ -a_- \sigma^1 & a_+ \end{pmatrix}$$

We can write this as:

$$S^{-1} = a_+ - a_- \gamma_0 \gamma_1$$

Exercise 22: (*) Verify that $SS^{-1} = 1$. $\square$

Exercise 23: (*) Show that:

$$\delta_\alpha^{\omega'} S^{-1} \gamma^\alpha S = \Lambda_\mu^{\omega'} \gamma^\mu$$

Note that the δ function is just an artifact from our primed indices notation: it is an identity matrix whenever we calculate anything. So, for example, you will have to show that:

$$S^{-1} \gamma^0 S = \gamma \gamma^0 - \gamma |\beta| \gamma^3$$

which you will do showing:

$$\begin{pmatrix} a_+ & -a_- \sigma^1 \\ -a_- \sigma^1 & a_+ \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \begin{pmatrix} a_+ & a_- \sigma^1 \\ a_- \sigma^1 & a_+ \end{pmatrix} = \gamma \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} - \gamma |\beta| \begin{pmatrix} 0 & \sigma^1 \\ -\sigma^1 & 0 \end{pmatrix}$$

You'll have related calculations for $\gamma^1, \gamma^2, \gamma^3$ as well. Tip: remember when grinding through algebra to simplify your notation. I suggest:

$$a_+ \rightarrow a, \quad a_- \rightarrow b, \quad \sigma^1 \rightarrow \sigma$$

$\square$

Exercise 24: (*) Show that:

$$S^\dagger \gamma^0 S = \gamma^0$$

$\square$

Exercise 25: Notice that our original solutions to the Dirac equations are simply plane wave solutions with $\vec{p} = 0$ and

$$u^{(1)} \equiv \begin{pmatrix} 1 \\ 0 \\ 0 \\ 0 \end{pmatrix}, \quad u^{(2)} \equiv \begin{pmatrix} 1 \\ 0 \\ 0 \\ 0 \end{pmatrix}$$

calculate $Su^{(1)}$ and $Su^{(2)}$. Compare to our results for $u^{(1)}$ and $u^{(2)}$ from the previous solution. Hint: Don't get confused by an unexpected minus sign in your answer... if you boost to a frame with speed $\beta > 0$ in the x direction, then a particle at rest in the original frame will appear to have momentum $-\gamma\beta mc$ in the new frame! $\square$

14 Bilinear Covariants

Suppose we want to construct a Lorentz invariant scalar from the bi-spinor ψ . It would be quite reasonable to first consider:

$$\psi^\dagger \psi = \begin{pmatrix} \psi_1^* & \psi_2^* & \psi_3^* & \psi_4^* \end{pmatrix} \begin{pmatrix} \psi_1 \\ \psi_2 \\ \psi_3 \\ \psi_4 \end{pmatrix} = |\psi_1|^2 + |\psi_2|^2 + |\psi_3|^2 + |\psi_4|^2$$

which is certainly a scalar, but is it Lorentz invariant? We can check:

$$(\psi^\dagger \psi)' = (\psi^\dagger)' (\psi)' = (S\psi)^\dagger (S\psi) = \psi^\dagger S^\dagger S \psi$$

which would be Lorentz invariant if $S^\dagger S = 1$ (i.e. if S is unitary). But alas, S is not unitary, as you will show in an exercise.

Exercise 26: Show that S is not unitary for $\beta > 0$. First note that $S^\dagger = S$ and then calculate $S^\dagger S = S^2$. $\square$

So instead consider the quantity: $\psi^\dagger \gamma^0 \psi$ we have:

$$(\psi^\dagger \gamma^0 \psi)' = (\psi^\dagger)' \gamma^0 (\psi)' = (S\psi)^\dagger \gamma^0 (S\psi) = \psi^\dagger S^\dagger \gamma^0 S \psi$$

However, you showed in a previous exercise that:

$$S^\dagger \gamma^0 S = \gamma^0$$

which leads to:

$$(\psi^\dagger \gamma^0 \psi)' = \psi^\dagger \gamma^0 \psi$$

which shows that this is indeed a Lorentz invariant scalar. To clean up our notation, let's define the Dirac adjoint of a bi-spinor as:

$$\bar{\psi} \equiv \psi^\dagger \gamma^0$$

We've just seen that $\bar{\psi} \psi$ is a Lorentz invariant scalar.

Next let's see how

$$\bar{\psi} \gamma^\mu \psi$$

transforms. In the prime frame, using primed indices, we have the quantity:

$$\bar{\psi}' \gamma^{\omega'} \psi'$$

but the gamma matrices are the same in every frame, so we have:

$$\begin{aligned} (\bar{\psi})' \gamma^{\omega'} (\psi)' &= \delta_\mu^{\omega'} (\psi^\dagger)' \gamma^0 \gamma^\mu \psi' \\ &= \delta_\mu^{\omega'} \psi^\dagger S^\dagger \gamma^0 \gamma^\mu S \psi \\ &= \delta_\mu^{\omega'} \psi^\dagger S^\dagger \gamma^0 S S^{-1} \gamma^\mu S \psi \\ &= \delta_\mu^{\omega'} \psi^\dagger \gamma^0 S^{-1} \gamma^\mu S \psi \\ &= \bar{\psi} \delta_\mu^{\omega'} S^{-1} \gamma^\mu S \psi \end{aligned}$$

In a previous exercise, you showed that:

$$\delta_\mu^{\omega'} S^{-1} \gamma^\alpha S = \Lambda_\mu^{\omega'} \gamma^\mu$$

which leads to:

$$(\bar{\psi})' \gamma^{\omega'} (\psi)' = \Lambda_\mu^{\omega'} \bar{\psi} \gamma^\mu \psi \quad (15)$$

which shows that:

$$\bar{\psi} \gamma^\mu \psi$$

transforms as a four-vector.

15 Hermiticity of the Gamma Matrices

Any representation of the gamma group, must have:

$$(\gamma^0)^2 = 1$$

If we further impose that:

$$(\gamma^0)^\dagger = \gamma^0$$

then we see that γ^0 is unitary:

$$(\gamma^0)^\dagger \gamma^0 = 1$$

Furthermore we must have,

$$(\gamma^1)^2 = -1, (\gamma^2)^2 = -1, (\gamma^3)^2 = -1$$

so if we impose that:

$$(\gamma^1)^\dagger = -\gamma^1$$

then γ^1 is unitary by:

$$(\gamma^1)^\dagger \gamma^1 = -(\gamma^1)^2 = 1$$

and likewise for γ^2 and γ^3 . We can impose these four conditions as:

$$(\gamma^\mu)^\dagger = \gamma^0 \gamma^\mu \gamma^0$$

because e.g.

$$(\gamma^1)^\dagger = \gamma^0 \gamma^1 \gamma^0 = -\gamma^1 (\gamma^0)^2 = -\gamma^1$$

16 Orthornormality of Bispinors

With the Dirac adjoint operation defined, we can now tackle the orthogonality and completeness of the plane-wave solutions. Recall, that the particle solutions were of the form:

$$u^{(i)} = N \begin{pmatrix} u_A^{(i)} \\ \frac{\vec{\sigma} \cdot \vec{p}}{p^0 + m} u_A^{(i)} \end{pmatrix}$$

where N is a normalization factor we will soon determine, and we took:

$$u_A^{(1)} = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \quad u_A^{(2)} = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

We can now determine:

$$\begin{aligned}\bar{u}^{(i)} &= (u^{(i)})^\dagger \gamma^0 = N^* \left((u_A^{(i)})^\dagger \quad \frac{\vec{\sigma} \cdot \vec{p}}{p^0 + m} (u_A^{(i)})^\dagger \right) \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \\ \bar{u}^{(i)} &= N^* \left((u_A^{(i)})^\dagger \quad -\frac{\vec{\sigma} \cdot \vec{p}}{p^0 + m} (u_A^{(i)})^\dagger \right)\end{aligned}$$

We can finally calculate:

$$\bar{u}^{(i)} u^{(j)} = |N|^2 \left(1 - \frac{(\vec{p} \cdot \vec{\sigma})^2}{(p^0 + m)^2} \right) (u_A^{(i)})^\dagger u_A^{(j)}$$

Next recalling:

$$(\vec{p} \cdot \vec{\sigma})^2 = |\vec{p}|^2 = (p^0)^2 - (m)^2 = (p^0 + m)(p^0 - m)$$

We have:

$$\bar{u}^{(i)} u^{(j)} = |N|^2 \frac{2m}{p^0 + m} (u_A^{(i)})^\dagger u_A^{(j)}$$

Choosing:

$$N = \sqrt{p^0 + m} = \sqrt{\frac{E + m}{1}}$$

We have:

$$\bar{u}^{(i)} u^{(j)} = 2m (u_A^{(i)})^\dagger u_A^{(j)}$$

Now we arranged that:

$$(u_A^{(i)})^\dagger u_A^{(j)} = \delta_{ij}$$

so finally:

$$\bar{u}^{(i)} u^{(j)} = 2m \delta_{ij}$$

Which shows our solutions are orthogonal. For antiparticles, we find similarly:

$$\bar{w}^{(i)} w^{(j)} = 2m \delta_{ij}$$

17 Completeness of Bispinors

Remember in non-relativistic QM we wrote completeness like:

$$\sum |k\rangle \langle k| = 1$$

Our spinors $u_A^{(i)}$ are also complete, as their outer products sum to unity:

$$\begin{aligned}\sum_i u_A^{(i)} (u_A^{(i)})^\dagger &= \begin{pmatrix} 1 \\ 0 \end{pmatrix} \begin{pmatrix} 1 & 0 \end{pmatrix} + \begin{pmatrix} 0 \\ 1 \end{pmatrix} \begin{pmatrix} 0 & 1 \end{pmatrix} \\ &= \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} + \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix} \\ &= \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = I\end{aligned}$$

If we take an outer product of bispinors, we get a 4x4 matrix. So, using our results from the previous section:

$$\begin{aligned}
\sum_i u^{(i)} \bar{u}^{(i)} &= |N|^2 \begin{pmatrix} I & -\frac{\vec{p} \cdot \vec{\sigma}}{p^0 + m} \\ \frac{\vec{\sigma} \cdot \vec{p}}{p^0 + m} & -\frac{(\vec{\sigma} \cdot \vec{p})^2}{p^0 + m} \end{pmatrix} \\
&= \begin{pmatrix} (p^0 + m)I & -\vec{p} \cdot \vec{\sigma} \\ \vec{\sigma} \cdot \vec{p} & -(p^0 - m)I \end{pmatrix} \\
&= p^0 \begin{pmatrix} I & 0 \\ 0 & -I \end{pmatrix} - \begin{pmatrix} 0 & \vec{\sigma} \\ -\vec{\sigma} & 0 \end{pmatrix} \cdot \vec{p} + m \begin{pmatrix} I & 0 \\ 0 & I \end{pmatrix} \\
&= \gamma^\mu p_\mu + m \\
&= \not{p} + m
\end{aligned}$$

For antiparticles:

$$\sum_i w^{(i)} \bar{w}^{(i)} = \not{p} - m$$

18 Adjoint Form of Dirac Equation

The Dirac adjoint bi-spinor:

$$\bar{\psi} \equiv \psi^\dagger \gamma^0$$

satisfies an adjoint form of the Dirac equation. Starting from the Dirac equation:

$$\gamma^\mu \delta_\mu \psi = m \psi$$

and taking the “normal” adjoint of both sides:

$$\delta_\mu \psi^\dagger (\gamma^\mu)^\dagger = m \psi^\dagger$$

From the Hermiticity of the gamma matrices, we have:

$$(\gamma^\mu)^\dagger = \gamma^0 \gamma^\mu \gamma^0$$

so:

$$\delta_\mu \psi^\dagger \gamma^0 \gamma^\mu \gamma^0 = m \psi^\dagger$$

Multiplying both sides by γ^0 and using $(\gamma^0)^2 = 1$ so:

$$\delta_\mu \bar{\psi} \gamma^\mu = m \bar{\psi}$$

For the plane-wave particles, we have:

$$(\not{p} - m)u = 0, \quad \bar{u}(\not{p} - m) = 0$$

For the plane-wave antiparticles, we have:

$$(\not{p} + m)w = 0, \quad \bar{w}(\not{p} + m) = 0$$

19 Helicity Eigenstates

We saw that plane-wave solutions to the Dirac equation are only eigenstates of S_z when $p_x = p_y = 0$. More generally, plane-wave solutions to the Dirac equation are not eigenstates of spin:

$$[\vec{S}]_D = \frac{\hbar}{2} \begin{pmatrix} \vec{\sigma} & 0 \\ 0 & -\vec{\sigma} \end{pmatrix}$$

but they are eigenstates of the spin component in the direction of the momentum, which is known as the helicity:

$$\hat{h} = \frac{\vec{S} \cdot \vec{p}}{p}$$

In the Dirac representation, the helicity is:

$$[h]_D = \left[\frac{\vec{S} \cdot \vec{p}}{2p} \right]_D = \frac{\hbar}{2p} \begin{pmatrix} \vec{\sigma} \cdot \vec{p} & 0 \\ 0 & -\vec{\sigma} \cdot \vec{p} \end{pmatrix}$$

In the exercises, you will show that the helicity operator commutes with the Hamiltonian, so we can find plane-waves (eigenstates of the Hamiltonian) which are simultaneously eigenstates of helicity.

Exercise 27: (*) Show that:

$$[H, h] = 0$$

Work in the Dirac representation, where:

$$[H]_D = \begin{pmatrix} m & \vec{\sigma} \cdot \vec{p} \\ \vec{\sigma} \cdot \vec{p} & -m \end{pmatrix}$$

□

We wish to find plane-waves Ψ that are eigenstates of helicity:

$$\hat{h}\Psi = \lambda\Psi$$

or equivalently:

$$\frac{\hbar}{2p} \begin{pmatrix} \vec{\sigma} \cdot \vec{p} & 0 \\ 0 & -\vec{\sigma} \cdot \vec{p} \end{pmatrix} \begin{pmatrix} u_A \\ u_B \end{pmatrix} = \lambda \begin{pmatrix} u_A \\ u_B \end{pmatrix} \quad (16)$$

from which we infer:

$$(\vec{\sigma} \cdot \vec{p}) u_A = \frac{2p\lambda}{\hbar} u_A \quad (17)$$

In the exercises, you will show that the allowed eigenvalues are:

$$\lambda \pm \frac{\hbar}{2}$$

exactly as expected for a component of spin for a spin one-half particle.

Exercise 28: Using

$$(\vec{\sigma} \cdot \vec{p})^2 = p^2$$

verify that the eigenstates of helicity are

$$\lambda = \pm \frac{\hbar}{2}.$$

Hint: calculate

$$(\vec{\sigma} \cdot \vec{p})^2 u_A$$

by applying Equation 17 twice. $\square$

Starting from Equation 8:

$$u_B = \frac{\vec{\sigma} \cdot \vec{p}}{E + m} u_A$$

and then applying Equation 17 we have

$$u_B = \frac{2p\lambda/\hbar}{E + m} u_A \quad (18)$$

which can be used to determine u_B for a simultaneous eigenstate of helicity and the Hamiltonian once we have u_A for an eigenstate of helicity. It is convenient to work in spherical coordinates, where:

$$\left[\frac{\vec{\sigma} \cdot \vec{p}}{p} \right]_D = \frac{1}{p} \begin{pmatrix} p_z & p_x - ip_y \\ p_x - ip_y & -p_z \end{pmatrix} = \begin{pmatrix} \cos \theta & \sin \theta e^{-i\phi} \\ \sin \theta e^{i\phi} & -\cos \theta \end{pmatrix}$$

and setting:

$$u_A = \begin{pmatrix} a \\ b \end{pmatrix}$$

The eigenstate for $\lambda = \frac{\hbar}{2}$ will satisfy:

$$\begin{pmatrix} \cos \theta & \sin \theta e^{-i\phi} \\ \sin \theta e^{i\phi} & -\cos \theta \end{pmatrix} \begin{pmatrix} a \\ b \end{pmatrix} = \begin{pmatrix} a \\ b \end{pmatrix}$$

From which we infer:

$$a \cos \theta + \sin \theta e^{-i\phi} b = a$$

or:

$$\frac{b}{a} = \frac{1 - \cos \theta}{\sin \theta} e^{i\phi}$$

and using the half-angle identities:

$$\frac{b}{a} = \frac{\sin(\theta/2)}{\cos(\theta/2)} e^{i\phi}$$

Defining:

$$s = \sin(\theta/2)$$

and

$$c = \cos(\theta/2)$$

we see that for:

$$a = c$$

and

$$b = s e^{i\phi}$$

has

$$|a|^2 + |b|^2 = 1$$

Applying Equation 18 we determine the eigenstate for a particle with positive helicity:

$$u^\uparrow = \sqrt{E + m} \begin{pmatrix} c \\ s e^{i\phi} \\ \frac{p}{E+m} c \\ \frac{p}{E+m} s e^{i\phi} \end{pmatrix}$$

similarly, we can determine the eigenstate for a particle with negative helicity:

$$u^\downarrow = \sqrt{E+m} \begin{pmatrix} -s \\ ce^{i\phi} \\ \frac{p}{E+m}s \\ -\frac{p}{E+m}ce^{i\phi} \end{pmatrix}$$

The antiparticle eigenstates are:

$$v^\uparrow = \sqrt{E+m} \begin{pmatrix} \frac{p}{E+m}s \\ -\frac{p}{E+m}ce^{i\phi} \\ -s \\ ce^{i\phi} \end{pmatrix} \quad v^\downarrow = \sqrt{E+m} \begin{pmatrix} \frac{p}{E+m}c \\ \frac{p}{E+m}se^{i\phi} \\ c \\ se^{i\phi} \end{pmatrix}$$

In the ultra-relativistic limit:

$$E \gg m$$

these reduce to:

$$u^\uparrow = \sqrt{E} \begin{pmatrix} c \\ se^{i\phi} \\ c \\ se^{i\phi} \end{pmatrix} \quad u^\downarrow = \sqrt{E} \begin{pmatrix} -s \\ ce^{i\phi} \\ s \\ -ce^{i\phi} \end{pmatrix}$$

and

$$v^\uparrow = \sqrt{E} \begin{pmatrix} s \\ -ce^{i\phi} \\ -s \\ ce^{i\phi} \end{pmatrix} \quad v^\downarrow = \sqrt{E} \begin{pmatrix} c \\ se^{i\phi} \\ c \\ se^{i\phi} \end{pmatrix}$$

Exercise 29: Our derivation did not use the second constraint from Equation 16 on u_B . Show that this is already satisfied by our solutions, which are simultaneous Eigenstates of the Hamiltonian.